Question	Answer	Marks	Partial Marks
7(b)	67.8 or 67.9 or 67.83 to 67.88	**3**	**M2** for [$\sin B =$] $\frac{115\times\sin 72}{118.1}$ oe or **M1** for $\frac{115}{\sin B}=\frac{118.1}{\sin 72}$ oe
7(c)(i)	255	**3**	**B1** for bearing of *B* from *A* is 75 soi **M1** for 180 + 75 oe
7(c)(ii)	[00]7.2	**2**	**M1** for *their* **(c)(i)** – *their* **(b)** –180
7(d)	11.8 or 11.82 to 11.83	**3**	**M1** for 115 ÷ 35 oe **M1** for *their* speed in m/s × 60 × 60 ÷ 1000
7(e)	76.1 or 76.08 to 76.09	**3**	**M2** for $\frac{\text{distance}}{80}=\sin 72$ oe or **M1** for distance required is perpendicular to *AC* soi
8(a)(i)	Correct sketch	**2**	**B1** for correct shape but inaccurate
8(a)(ii)	Rotational [symmetry] order 2 [centre] (180, 0)	**2**	**B1** for rotational [symmetry]
8(b)	48.6 or 48.59 to 48.60 and 131.4 or 131.40 to 131.41	**3**	**B2** for 48.6 or 48.59 to 48.60 or 131.4 or 131.40 to 131.41 or **M1** for sin *x* = 0.75 or better If 0 scored, **SC1** for two answers adding to 180
8(c)(i)	$(x + 5)^2 – 11$	**2**	**M1** for $(x + 5)^2 + k$ or $(x + \textit{their } 5)^2 + 14 – (\textit{their } 5)^2$ or $a = 5$
8(c)(ii)	Sketch of U-shaped parabola with a minimum indicated at (–5, –11) with no part of graph in 4th quadrant	**3**	**FT** *their* $(x + 5)^2 – 11$ provided in that form **B1** for U shape curve **B1FT** for turning point at (–5, *k*) or (*k*, –11)
9(a)	39[.0] or 39.03 to 39.04…	**3**	**M2** for $\frac{165}{360}\times 2\times\pi\times 8 + 16$ or **M1** for $\frac{165}{360}\times 2\times\pi\times 8$
9(b)	2.71 or 2.708…	**4**	**M3** for $\sqrt{\frac{\frac{165}{360}[\times\pi]\times 8^2}{4[\times\pi]}}$ oe or **M2** for $r^2 = \frac{\frac{165}{360}[\times\pi]\times 8^2}{4[\times\pi]}$ oe or **M1** for $\frac{165}{360}\times\pi\times 8^2$ oe seen

Question	Answer	Marks	Partial Marks
9(c)(i)	3.67 or 3.666 to 3.667	**2**	**M1** for $\frac{165}{360}\times 2[\times\pi]\times 8 = 2[\times\pi]\times r$ or better or for $\frac{165}{360}[\times\pi]\times 8^2 = [\pi\times]r\times 8$ or better
9(c)(ii)	100 or 100.0 to 100.1… final answer	**4**	**M3** for $\frac{1}{3}\pi\times their(c)(i)^2\times\sqrt{8^2 - their\ \text{radius}^2}$ or **M2** for $\sqrt{8^2 - their\ \text{radius}^2}$ or **M1** for $(their\ (c)(i))^2 + h^2 = 8^2$
10(a)(i)	15.7 or 15.65...	**3**	**M2** for $\sqrt{(4-10)^2+(4--3)^2}$ oe or **M1** for $(-4-10)^2 + (4--3)^2$ oe
10(a)(ii)	$\frac{-10-4}{4--3}$ [= –2] oe	**M1**	
	10 = –2(–3) + c Or –4 = –2(4) + c and correct completion to $y = -2x + 4$	**A1**	
10(a)(iii)	$y = \frac{1}{2}x + \frac{11}{4}$ oe	**4**	**M1** for grad = ½ soi **M1** for [midpoint =] (½, 3) **M1** for substitution of (1/2, 3) into *their* $y = mx + c$ oe
10(b)(i)	$\left(-\frac{1}{3}, -\frac{22}{27}\right)$ oe and (–5, 50)	**6**	**B2** for $3x^2 + 16x + 5$ Or **B1** for one correct **M1** for derivative = 0 or *their* derivative = 0 **M1** for [x =] $-\frac{1}{3}$ and [x =] –5 **B1** for $-\frac{22}{27}$ and 50
10(b)(ii)	$\left(-\frac{1}{3}, -\frac{22}{27}\right)$ minimum (–5, 50) maximum with correct reasons	**3**	**B2** for one correct with reason or **M1** for correct attempt e.g. 2nd derivatives, gradients or sketching